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Review

Generative artificial intelligence for sustainable development: predictive trend analysis in key sectors using natural language processing

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Abstract
Generative Artificial Intelligence (GenAI) has revolutionized multiple industries by improving efficiency and promoting innovation in fields such as business, education, healthcare, and cybersecurity. This study evaluates the transformative impact of GenAI on advancing the United Nations' Sustainable Development Goals (SDGs) by analyzing research trends and applications. In this research, authors utilized Latent Dirichlet Allocation (LDA), a Natural Language Processing (NLP) technique. The work delineates emergent themes and sector-specific progressions in GenAI research. The dataset consists of 2162 research articles published from 2020 to 2025, obtained from the Scopus database. The abstracts of these publications provide the foundation for analysis, facilitating a thorough examination of literature. The authors provided 10 topics, which are recent trends that future researchers can explore. The results indicate GenAI's capability in processing automation, creative enhancement, and innovation promotion while addressing ethical concerns such as prejudice, privacy, and social effects. The study indicates potential directions for ethical technology adoption by analyzing trends in GenAI applications. This work highlights the essential requirement for interdisciplinary methods and ethical frameworks to optimize GenAI's contributions to innovation while assuring alignment with sustainable and equitable development goals.

Keywords Generative artificial intelligence · Latent Dirichlet allocation (LDA) · Natural language processing (NLP) · Ethical AI implementation · Interdisciplinary applications · Sustainable development goals

1 Introduction

The emergence of Artificial Intelligence (AI) in recent years has been profoundly transformational, with technology infiltrating numerous areas and revolutionizing enterprises worldwide [1]. A study from McKinsey & Company says that by 2030, AI is expected to add $13 trillion to the economy. GenAI will play a big role in this growth by changing fields like content creation, healthcare diagnostics, and personalized marketing [2]. A Gartner survey forecasts that by 2025, 30% of outbound marketing communications from large enterprises will be synthetically generated [3]. These statistics illustrate the rapid proliferation of GenAI and its profound impact on business and society. GenAI is a notably disruptive field that may produce new content, including writing, graphics, music, and code, using patterns acquired from data [4].

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GenAI technologies, propelled by advanced machine learning algorithms and extensive neural networks, initiate a novel era of innovation across many sectors, including healthcare, finance, education, and entertainment [5].

The strength of GenAI lies in its ability to learn from extensive datasets and generate original, unlabeled data. This sharply contrasts with traditional AI systems, whose primary function is to analyze and predict based on historical data. Technologies such as Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Transformer-based models like GPT (Generative Pretrained Transformers) have opened up new possibilities for automation, creativity, and personalization [6]. Natural Language Processing (NLP), a field of artificial intelligence stressing the interface between computers and human language, is fundamental to this revolutionary potential. Natural language processing has recently made notable advancements, particularly in developing generative models that evaluate and generate human-like writing [7]. Using deep learning architectures, technologies such as GPT-3 and GPT-4 have shown amazing accuracy in natural language understanding, generation, and translation [8]. NLP models can produce coherent and contextually relevant text, influencing consumer service, content creation, law, education, and healthcare. Combining generative artificial intelligence with natural language processing has given companies, researchers, and practitioners an unmatched opportunity to look at how these technologies might affect many sectors [9]. The capacity to automate occupations, improve decision-making, and produce highly personalized solutions is revolutionizing organizational operations, competition, and innovation. Particularly regarding bias, privacy, and employment displacement, the rapid development of generative artificial intelligence and natural language processing has spurred questions about their ethical consequences, social repercussions, and possible risks [10].

Despite the widespread recognition of these technologies' potential, a more systematic approach to monitoring and predicting the trajectory of research and application is required. This is the domain where advanced data analytics and machine learning methodologies are applied. This research employs Natural Language Processing (NLP) methodologies to examine anticipated trends across many industries, thereby elucidating the transformative potential of generative artificial intelligence. The advancements in generative artificial intelligence and natural language processing research will be analyzed using the Latent Dirichlet Allocation (LDA) method [11]. Analyzing significant research in the domain reveals numerous breakthroughs and ongoing issues. For example, one research study referenced Generative Adversarial Networks (GANs), which have markedly enhanced AI-generated content by augmenting realism and precision [12]. In another study, the researchers built Transformer-based models such as GPT, which have transformed natural language processing (NLP) tasks, attaining state-of-the-art performance in text generation, summarization, and conversational AI [13]. Nonetheless, the existing research fails to thoroughly examine the adoption of these technologies across diverse industries and the evolution of their possible ethical and operational difficulties over time. In contrast to other NLP methodologies like Non-Negative Matrix Factorization (NMF) and deep learning-based topic models, LDA provides superior interpretability, scalability, and computational efficiency for managing extensive academic datasets. Nonetheless, deep learning techniques, albeit yielding superior accuracy, frequently exhibit a deficiency in transparency and necessitate substantial computer resources [14].

This paper presents a detailed taxonomy of the current landscape of GenAI research, obtained from Latent Dirichlet Allocation (LDA)-based topic modeling applied to a corpus of 2162 Scopus-indexed publications. The taxonomy categorizes GenAI research into eleven thematic domains: Process Automation, Creative Arts, Healthcare Applications, Educational Technology, Business Optimization, Cybersecurity, Content Creation, Data Science, and Ethics and Bias. Each domain corresponds to one or more pertinent United Nations Sustainable Development Goals (SDGs), underscoring GenAI's potential contributions to global development objectives. For instance, GenAI applications in healthcare immediately contribute to SDG 3 (Good Health and Well-Being), whilst advancements in educational technology correspond with SDG 4 (Quality Education). The incorporation of subjects like cybersecurity and ethical AI highlights the interdisciplinary character and societal implications of GenAI research. This taxonomy integrates fragmented research initiatives into organized categories and functions as a strategic framework for future research, policymaking, and the responsible implementation of technology to promote sustainable and equitable innovation. Research taxonomy of the study represented in Fig. 1.

This research thoroughly examines the growing trends and emerging patterns in literature, providing significant insights into these subjects' current status and future trajectories. This study utilizes LDA, adept at revealing latent topics in extensive datasets, to identify significant research areas, monitor their evolution over time, and offer predictive insights regarding the impact of GenAI and natural language processing across diverse sectors. LDA is an effective topic modeling approach that identifies latent topics within a collection of publications [15]. It is ideal for uncovering trends and patterns in extensive datasets such as academic research articles. In the context of this research, LDA will be used to analyze a corpus of research papers related to GenAI and NLP extracted from the Scopus database [16]. By applying LDA

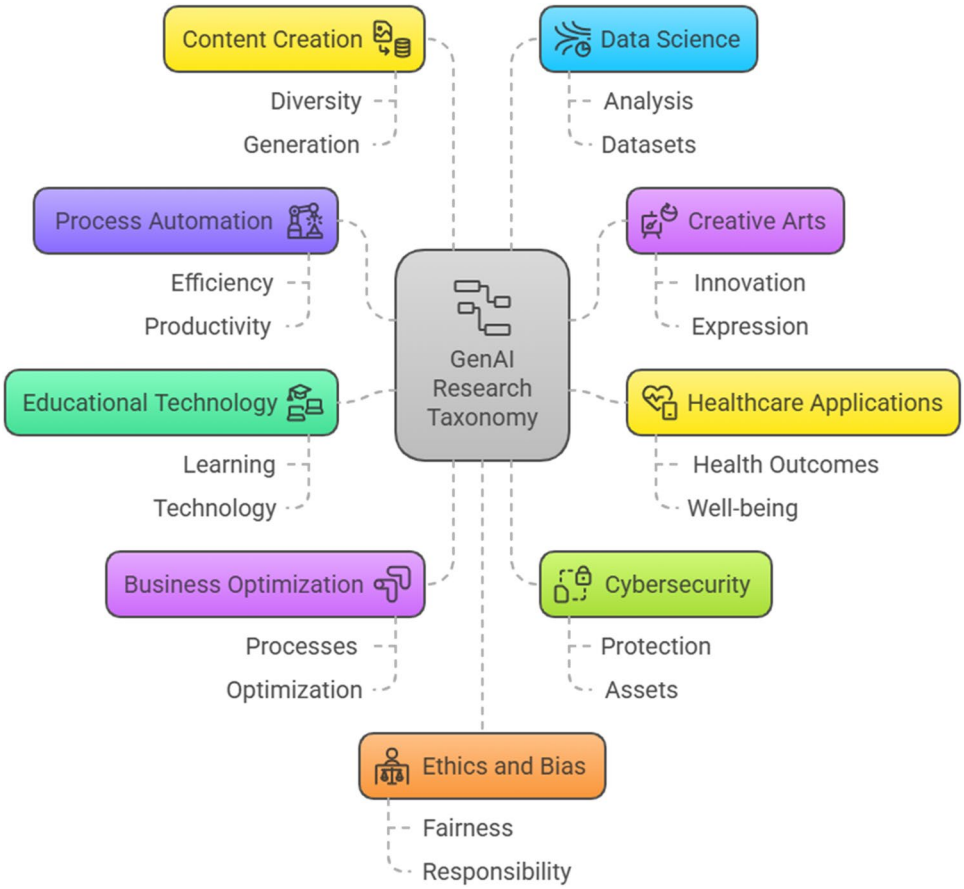


Fig. 1 Proposed taxonomy of GenAI research themes

to this corpus, researchers can identify key topics and trends within the literature, track how these trends have evolved, and predict future directions for research in GenAI and NLP. Using LDA enables the extraction of valuable insights from vast amounts of text, providing a clear picture of where the field is heading and how GenAI is being applied in different sectors [17–20]. To guide this study, the following research questions are proposed:

RQ1: What are the key research trends and emerging themes in GenAI across different sectors?

RQ2: How have these trends evolved based on available literature?

RQ3: How can future research address the identified gaps and leverage AI advancements more effectively?

1.1 The motivation for the research

The main topics of this research study are the expansion of GenAI research, its multidisciplinary influence, ethical and societal issues, and the evolution of predictive insights. Though there is no thorough study of trends and paths, the fast development of artificial intelligence technologies has resulted in major contributions in many spheres. This work investigates trends utilizing cutting-edge techniques, including latent Dirichlet Allocation (LDA). It also emphasizes the importance of a methodical approach in comprehending the ethical consequences and effects of generative artificial intelligence on society, such as prejudice, privacy, and job displacement. The study also investigates the junction of generative artificial intelligence with natural language processing (NLP), providing unheard-of chances for automation, content creation, and decision-making. The study underlines the need to fill inefficiencies, encourage creativity, and open the path for moral and significant uses of generative artificial intelligence [21, 22].

This article is organized into five principal sections to thoroughly examine the transformational potential of GenAI across multiple sectors. The introduction establishes the framework by delineating the study's scope, objectives, and importance, highlighting the necessity for predictive insights into developing trends. The Literature Review examines the historical development of GenAI, emphasizing its applicability in healthcare, education, and business while pinpointing current research deficiencies. The Research Methodology outlines the application of Latent Dirichlet Allocation (LDA) for topic modeling, specifying the dataset sourced from the Scopus database and the analytical tools utilized. The Results

and Discussion delineate significant findings about sector-specific patterns and their ramifications while addressing ethical concerns such as bias and privacy and recognizing the study's limitations. The Conclusion and Future Scope encapsulate the discoveries, provide directions for additional research, and emphasize the practical ramifications of GenAI in influencing creativity and decision-making across several fields.

1.2 Key contributions of the research

- Conducting a large-scale trend analysis using 2162 research abstracts from the Scopus database (2020–2025), enabling the identification of key research themes in GenAI through topic modeling.
- Applying Latent Dirichlet Allocation (LDA) to uncover hidden research clusters and map them to relevant SDGs, thereby offering a structured and interpretable framework for future research and policy design.
- Highlighting interdisciplinary applications and ethical considerations, the study offers predictive insights into how GenAI can responsibly shape innovation across sectors.

This contribution is positioned at the intersection of AI, sustainable development, and digital ethics. It aims to guide researchers, practitioners, and policymakers in understanding the evolving landscape of GenAI while promoting responsible adoption aligned with global goals. This paper is structured to explore the transformative role of Generative Artificial Intelligence (AI) in achieving Sustainable Development Goals (SDGs) through predictive trend analysis using Natural Language Processing (NLP) techniques. The Introduction outlines the rapid rise of GenAI, its relevance to SDGs, and the need for predictive insights, followed by the Literature Review, which traces the historical evolution and sector-specific applications of GenAI while identifying existing research gaps. The Research Methodology details the use of Latent Dirichlet Allocation (LDA) on a dataset of 2162 Scopus-indexed articles, describing the preprocessing steps and justification for choosing LDA. In the Implementation and Results section, the paper presents trends in publications, key contributors, global collaborations, and identifies ten major research topics, each mapped to relevant SDGs. Finally, the Conclusion and Future Scope highlights the potential of GenAI to drive innovation across sectors, underlines the importance of ethical and interdisciplinary approaches, and recommends future research directions, including broader data integration and advanced modeling techniques.

2 Literature review

GenAI extends back to the 1950s when key concepts like machine learning and neural networks were initially created [23]. The Dartmouth Workshop 1956 defined AI as a study field, providing the framework for generative technologies [24]. By the 1980s, rule-based systems and statistical models enabled early generative techniques. The 2000s witnessed the introduction of deep learning models such as Restricted Boltzmann Machines and Variational Autoencoders, leading to Ian Goodfellow's pioneering Generative Adversarial Networks (GANs) in 2014, which changed image, video, and audio synthesis [25]. The late 2010s heralded a paradigm change with transformer models like OpenAI's GPT and Google's BERT, advancing text creation [26]. By 2020, large-scale models like GPT-3 displayed unparalleled natural language interpretation and creation skills, while diffusion models like DALL·E 2 pushed the bounds of visual generation. GenAI became a widespread tool for content production, gaming, and virtual assistants, growing further with ethical considerations and multimodal capabilities for varied applications. Today, it continues to alter industries with real-time, hyper-realistic creative capacities. Figure 2 shows how GenAI evolved and where it is right now. GenAI is being applied across various sectors, each benefiting from its unique capabilities. Generative models are used in some areas of healthcare, including synthetic medical data production, the design of new medicinal substances, and the improvement of medical imaging [27]. Generative artificial intelligence is used in customer service augmentation via chatbots and virtual assistants, fraud detection, and trading strategy optimization [28]. While AI-powered solutions in education are automating administrative labor and generating tailored learning experiences, GenAI is used in the entertainment sector to develop realistic virtual environments, produce content, and help with scriptwriting and music creation [29]. Generative Artificial Intelligence (GenAI) has emerged as a transformative force across multiple industries, reshaping workflows, automating tasks, and fostering innovation at an unprecedented scale. These industries include education [30], higher education [31], e-commerce [32], and many more. The transformative

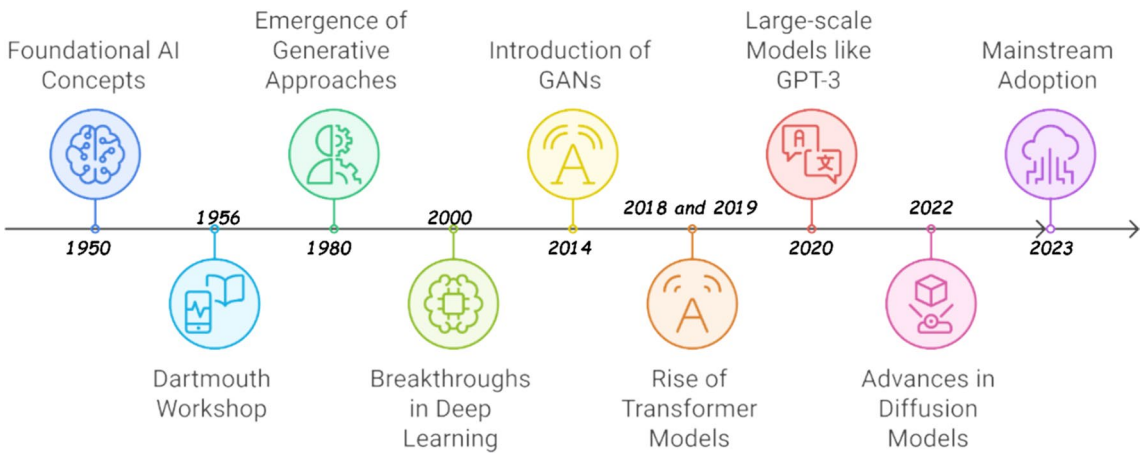


Fig. 2 Evolution of GenAI

power of GenAI, particularly when combined with Natural Language Processing, is reshaping industries and driving innovation across the globe.

One of the research studies published in 2023 claimed that integrating generative artificial intelligence (AI) in education has sparked a paradigm shift in teaching and learning, offering unprecedented opportunities and significant challenges [33]. This paper examines the ethical implications and equity issues related to the application of AI in education, highlighting the necessity for a deliberate, ethical, and inclusive incorporation of these technologies. It examines the complex function of AI in scientific publishing, credit attribution, and professional environments. Another study examined the application of GenAI in content production within the entertainment sector, specifically in cinema, music, and video games [34]. This research emphasized the growing utilization of AI in script generation, revolutionizing conventional storytelling by facilitating the development of plotlines, conversations, and characters that resonate with viewer preferences. A study on GenAI in marketing and advertising demonstrates its use in producing tailored content and customizing advertisements for targeted consumers [35]. It emphasizes its capacity to derive client insights and formulate dynamic social media interaction tactics, revolutionizing marketing techniques, enhancing campaign efficiency, and cultivating better customer-brand ties. A few studies have examined the role of GenAI in retail and e-commerce, particularly its applications in product design, personalized recommendations, virtual try-ons, and dynamic pricing models. One of the research studies in this domain presented an innovative mobile application that integrates sustainable e-commerce, AI-driven personalized recommendations, and efficient warranty management [36]. The research studies investigated consumer behavior, attitudes about pre-owned products, and opinions on sustainability utilizing secondary research, primary research, development, and usability testing combined. The researchers have also focused on investigating and exploring the use of GenAI in the industry [37]. To increase efficiency, the industrial sectors in this field are developing prototypes, simulating manufacturing techniques, and improving product design [38]. This work aims to decipher the influence of generative artificial intelligence across several domains and offer a road map for the next research and implementation by using cutting-edge data analytics methods, including LDA. Utilizing research data analysis from the Scopus database, the researchers hope to expose the main trends, difficulties, and prospects defining the direction of generative artificial intelligence and natural language processing. The historical progression of GenAI reveals significant advancements, including the creation of Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and extensive transformer-based models such as GPT-3 and GPT-4, while also emphasizing deficiencies in comprehending the ramifications of these technologies across various sectors [39]. Despite considerable advancements in content production, picture synthesis, and natural language processing, the literature frequently exhibits a deficiency in rigorous examination of sector-specific applications and their ethical ramifications. Furthermore, although models like GPT-4 exhibit exceptional proficiency in producing human-like text, the shift from technological advancement to practical implementation uncovers numerous persistent challenges, especially in confronting ethical issues such as bias, privacy, and job displacement. Current study has predominantly concentrated on technological advancement, neglecting essential inquiries regarding practical implementation, regulatory structures, and multidisciplinary integration.

This study applies a systematic, data-oriented methodology to examine evolving patterns and trends in GenAI across several areas. This study employs Latent Dirichlet Allocation (LDA) on an extensive dataset of 2,162 research papers to achieve the following objectives:

1. Identifying sector-specific applications and research gaps in GenAI.
2. Examining the ethical considerations and societal implications of its adoption.
3. Providing a predictive framework for understanding how GenAI will shape future innovation and decision-making across industries.

Although prior reviews have examined the emergence of GenAI, the majority are either descriptive or confined to specific sectors, failing to provide a comprehensive, data-driven examination across various fields. This study presents a novel contribution by utilizing Latent Dirichlet Allocation (LDA) topic modeling on a curated dataset of 2162 Scopus-indexed articles, facilitating the discovery of developing subject clusters and their direct correlation with certain Sustainable Development Goals (SDGs). Our methodology, in contrast to conventional reviews, reveals hidden research trends, delineates ethical and practical issues, and emphasizes underexamined areas, so offering a predictive perspective for future research and policy development in GenAI. The study integrates SDG mapping with topic modeling, employs quantitative methods in qualitative research, and provides a visual taxonomy framework for a logical field structure overview which is rare in existing literature.

3 Research methodology

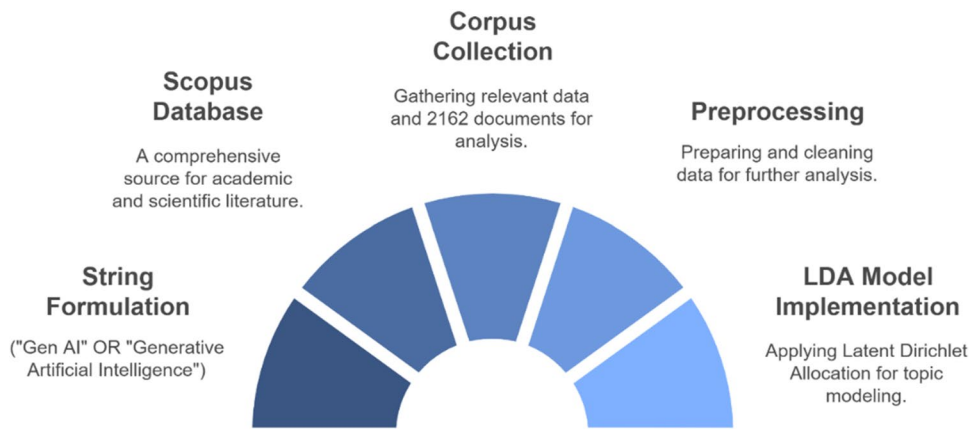
NLP stands for Natural Language Processing and is a subfield of Artificial Intelligence that deals with building systems designed to process natural language [40]. It drives the creation of narratives, or their understanding, by allowing machines to interpret text or speech. NLP is thus an interface that brings together computational linguistics and elements of machine learning to translate natural language between people and machines [41]. Major areas of using NLP are opinion mining, translation, text generation and summarization, speech recognition, and conversational agents. Some implicated operations include tokens, stemming, lemmatization that reduces tokens to root form, named entity recognition that recognizes the occurrence of important entities, and syntactic parser that transforms text data to structured data [42, 43]. One of the most complicated methods of the NLP is topic modeling – the art of identifying abstract topics within a certain number of documents. The algorithms scrutinize the words in the documents and set out patterns or relationships in which similar words are grouped in topics. Common topic modeling techniques include:

- Latent Semantic Analysis (LSA): Singular value decomposition determines the relations between terms and documents in vector space models [44].
- Non-negative Matrix Factorization (NMF): A method that uses the properties of linear algebra to factorize and extract low-level representation from the document-term matrix with the discovery of latent topical structure [45].
- Latent Dirichlet Allocation (LDA): A generative model for unsupervised learning of texts that assumes each document is a composition of topics, which are distributions of words [16, 46].

Of these methods, LDA is considered better because, inter alia, it is probabilistic and easily explained. LDA assumes that documents are generated by a hidden probabilistic process, where each document includes a set of themes, and Each theme can be described by a distribution of words [47]. This probabilistic approach also makes LDA flexible for large and heterogeneous datasets because it can capture the semantic relationship between words and topics well [48, 49]. The research methodology of the study is represented in Fig. 3.

Furthermore, in LDA, it is easy to determine the weights of a specific topic in a document, which is useful in other related applications such as content recommendation, document clustering, and information retrieval. The favorable attributes of LDA include scalability, sparsity of the co-occurrence data, and the interpretational capability of the themes. Compared to the deterministic LSA, it is more capable of capturing semantic details. Also, the non-negativity constraint is handled more naturally than by NMF, making it the go-to method for real-world topic modeling [50, 51].

Fig. 3 Research methodology



3.1 Dataset information and preprocessing

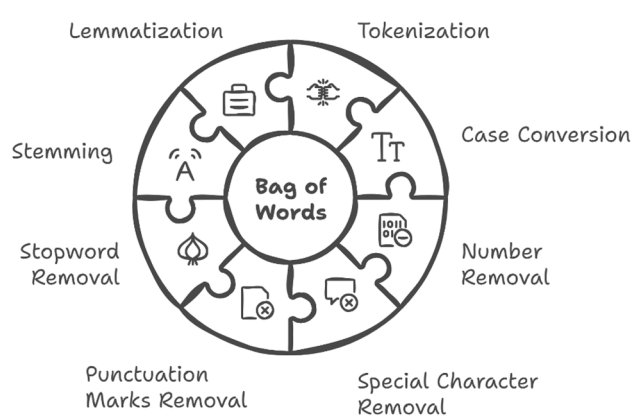
Various online databases host published articles, but Scopus is the most reliable and extensive database for scholarly works [52], and various researchers used it in their studies. The Scopus database was selected for experimentation in this study, influenced by prior state-of-the-art research. The authors identified search phrases "GenAI" and "Generative Artificial Intelligence", which were combined into the search string:

(TITLE (("GenAI" OR "generative artificial intelligence")))

This query initially retrieved 2445 articles. After applying a language filter to include only English articles. Following the inclusion criteria, which considered articles from journals, conferences, and book chapters, a final set of 2162 articles was selected for experimentation. The authors included all studies published in journals, conferences, and book chapters as authors predicting the recent trends, and for that, diversity from various platforms is required for analysis. The authors use abstracts of the published articles to perform the experiment as abstracts contain the study's summary [15]. The organized corpus was structured in an Excel file for further processing, facilitating a systematic data exploration and analysis approach. The preprocessing process is depicted in Fig. 4.

The author used multiple preprocessing actions during the research, which included removing stop words and numbers and stripping punctuation and special characters before performing stemming. Executing preprocessing techniques remains essential to extract an effective corpus for analysis [53]. The main purpose of these procedures is to target the removal of unhelpful information because random elements in data create noise that can detract from the effectiveness of the analysis. Noise contamination in the data analysis can yield false information and invalid conclusions [16]. Extended processing time occurs due to the complexity of maintaining nonessential dataset elements. The data quality improves when these preprocessing conditions allow researchers to work with focused and purified data that supports accurate and efficient analytical methods.

Fig. 4 Preprocessing process



3.2 Latent Dirichlet allocation (LDA)

LDA is used in topic modeling to identify hidden topics in documents. Latent Dirichlet Allocation (LDA) is a statistical probabilistic model [54]. It presumes that each document is a hybrid of various themes, and the distribution of words conditions all the themes. LDA functions by discovering probability distributions of words and documents concerning topics from the document collection. This process lets us obtain sensible topics describing the documents' content [54, 55]. Owing to the probabilistic LDA model, it is popular for handling voluminous and heterogeneous data to clearly understand the results and suitability for areas like document categorization, recommendations, and text clustering. Figure 5 represents the working of the LDA model [56, 57].

This experiment uses Python language through Google Colab to implement the LDA model on a preprocessed corpus. First, all the bigrams, trigrams, quadgrams, or n-grams are created and recognized with the help of Genism's model [58]. Further, these are removed from the data to improve the process as the individual interferes with no information [59, 60]; this experiment is mainly based upon the three input parameters, topics, and two hypermeters α and β. In the experiment, the value of α is considered 1/T, where T is the chosen number of topics [61], and the importance of β is constant to 0.01 for the selected topics. Therefore, the iteration used in this study for the experiment is 1000. This means the researchers have adjusted the model settings to ensure meaningful topic grouping. The topic count and related parameters were fine-tuned to produce the most consistent and interpretable results after multiple runs. No analysis provided the optimal number of topics for the corpus, but the authors achieved the highest coherence score of 0.62 for ten topics provided in the study [62]. Based on the researcher's suggestion, the author proposed 10 topics for the corpus of 2162 articles in this study. After processing, the dictionary of words is created, named Bag of Words (BOW). During the experiment, the iteration used for considered topic solution 10 is 1000, provided by Arun [63]. As a result, setting reasonable values for these parameters is essential, as they affect how often specific topics appear in the top results. LDA topic modeling processes the extracted bag of words (BOW), removing the most and least common words so the corpus is pure. The most frequent terms from the corpus with their frequency are shown in Fig. 6.

While Latent Dirichlet Allocation (LDA) offers scalability, transparency, and ease of interpretation, it also presents limitations in capturing deep contextual relationships and semantic coherence across documents. Recent advances such as Neural Topic Models (NTMs), BERTopic, and Contextualized Topic Models (CTMs) leverage transformer-based embeddings (e.g., BERT) to provide higher topic coherence and contextual relevance. However, these methods typically require more computational resources and are less interpretable, which can hinder their adoption in large-scale, cross-sectoral analyses such as this. Future research should explore hybrid approaches that balance interpretability with contextual depth.

From processing the corpus emerges the Bag of Words (BOW) model, which contains a dictionary of words. The dictionary contains isolated words that establish an analytical base. The authors implemented Latent Dirichlet Allocation (LDA) through these words to identify and measure the relevant most recent research areas needing further investigation [64]. The BOW approach enables LDA to identify hidden topics that provide meaningful insights about dominant areas in the data.

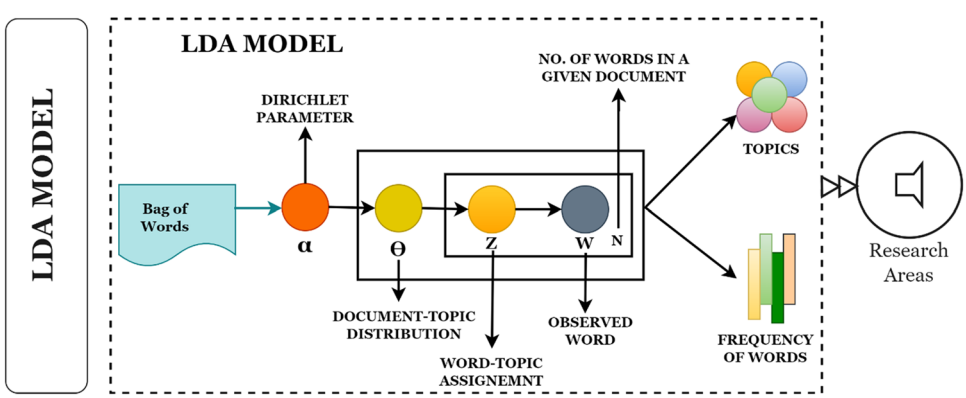


Fig. 5 Working of LDA model

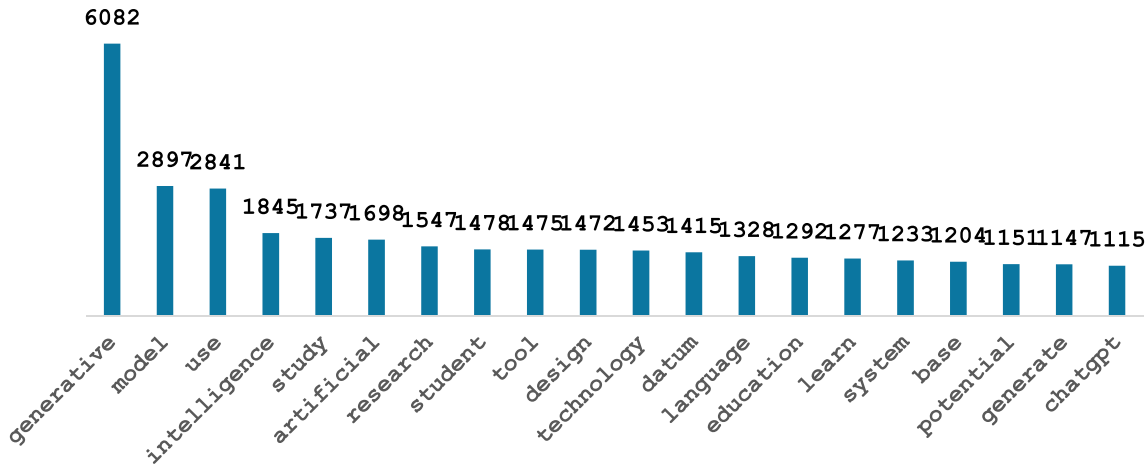


Fig. 6 Top 20 words with frequency

4 Implementation and results

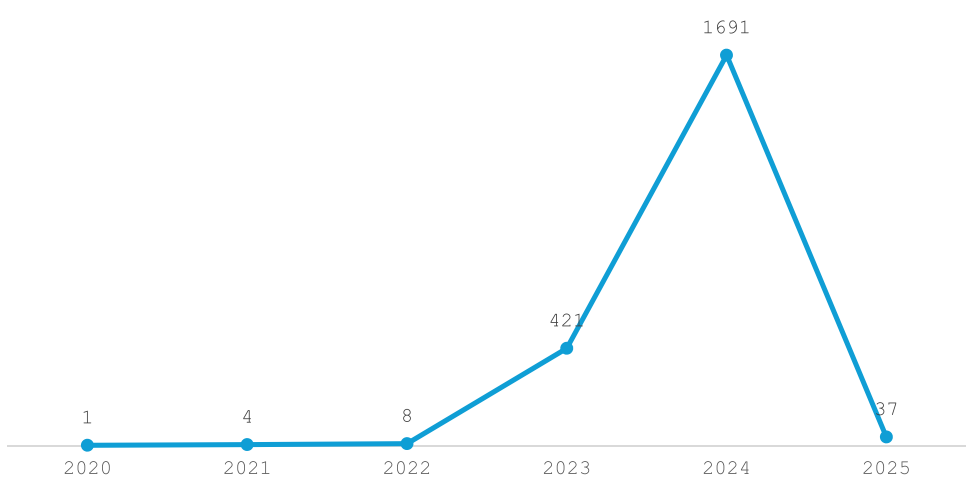
4.1 Year wise analysis

The study conducted a comprehensive year-wise analysis of 2162 published articles related to generative artificial intelligence on the Scopus database to understand research trends and growth trajectory. As shown in Fig. 7, the results show a sharp increase in publications in the past few years. In 2020, 1 paper was written about GenAI. In 2021, we saw 4 articles written; in 2022, there were 8 articles, and that's actually a slight increase compared to 2021. Nevertheless, 2023 saw remarkable growth as 421 articles were published, a dramatic rise in the research activity, and a full scholarly focus. In 2024, it intensified further with 1,691 articles, or 78 percent of total publications, demonstrating exponential progression, continuing importance, and pervasive field interest. Surprisingly, for 2025, there's a noticeable decline, with only 37 articles in the count. In some cases, the reason may be fully accounted for by data incompleteness for the year or other rising trends in the area of research. This analysis highlights the rapid adoption of GenAI and how it will set the direction for future technology while outlining areas for additional exploration in the future.

4.2 Journal analysis

Scholarly research dissemination and academic collaboration are performed via journals and conference proceedings, particularly in the nascent field of Generative Artificial Intelligence. In this study, the authors analyzed the publications that contribute the most to the top 10 publication sources in this domain concerning their counts (as shown in Fig. 8). It

Fig. 7 Yearwise growth of publication



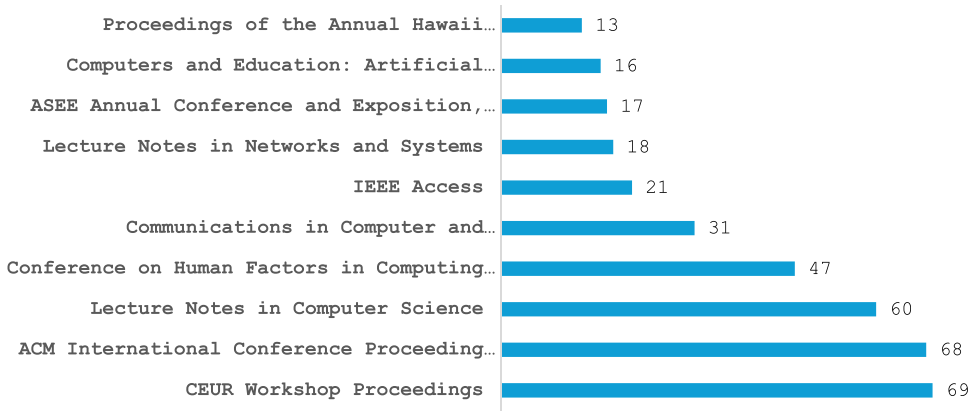


Fig. 8 Dominating source for publication

is heartening to see the CEUR Workshop Proceedings emerge as one of the leading sources, with 69 articles coming from its ranks, for it marks that working group as the most significant as it has acted as the place where cutting-edge developments and work done from workshop basis can be shared. Closely following, the 68 articles in the ACM International Conference Proceeding Series signal its crucial function in promoting the peer reviewing of conference contributions in the computing and AI realms. In all, 60 articles contributed to the Lecture Notes in Computer Science (LNCS) series, a renowned information resource for computational research on AI and generative models.

Moreover, 47 articles from the Conference on Human Factors in Computing Systems showcased the intersection between human–computer interaction and AI-driven technology. The Communications in Computer and Information Science rounded off with 31 articles for a heavy take on disseminating the AI research and further cemented its lifeblood among the two platforms. These sources indicate that GenAI is an area of research where researchers think and write about things far beyond the tech news cycles, exploring it from various venues to answer hard questions and push human capabilities in new directions.

4.3 Top authors analysis

Emphasizing their contributions and academic importance, Table 1 shows key author-level metrics for top researchers in generative artificial intelligence and the data collected as of December 2024. With 29 papers, 181 Scopus citations, and 89,419 Google Scholar citations, Niyato Dusit is the most active contributor. Reflecting on his great achievements, Michael Muller has a significant citation total of 201 on Scopus and 22,571 on Google Scholar. Each emerging researcher, Du, Hongyang, and Kang, Jiawen, has 20 papers; Kang has accumulated more citations on Google Scholar and Scopus. Early-stage contributor Hai-Jew, Shalin, has no data available for Google Scholar and the least significant imprint on document count and citations—3 on Scopus. With 14 papers, 119 Scopus citations, and 19,238 Google Scholar citations, eminent rising author Xiong Zehui suggests a growing influence that might indicate fundamental achievements. The study emphasizes the need to use several citation sites to evaluate academic influence fully.

The co-authorship network visualization in GenAI shows a highly collaborative research ecosystem, as represented in Fig. 9. In the middle, the central author, Niyato Dusit, exhibits his impact and multidisciplinary reach. The network also benefits other important authors, including Kang, Jiawen, and Du, Hongyang. The green cluster shows active subnetworks

Table 1 Academic impact of leading researchers in GenAI: a comparative analysis of document counts, citations, and indices

Author name	Document count (scopus)	Citation (scopus)	Citation (google scholar)	H-index (google scholar)	i-10 index (google scholar)
Niyato, Dusit	29	181	89,419	138	842
Du, Hongyang	20	138	3675	34	75
Kang, Jiawen	20	142	15,942	55	153
Xiong, Zehui	14	119	19,238	66	208
Hai-Jew, Shalin	11	3	NA	NA	NA
Muller, Michael	11	201	22,571	71	268

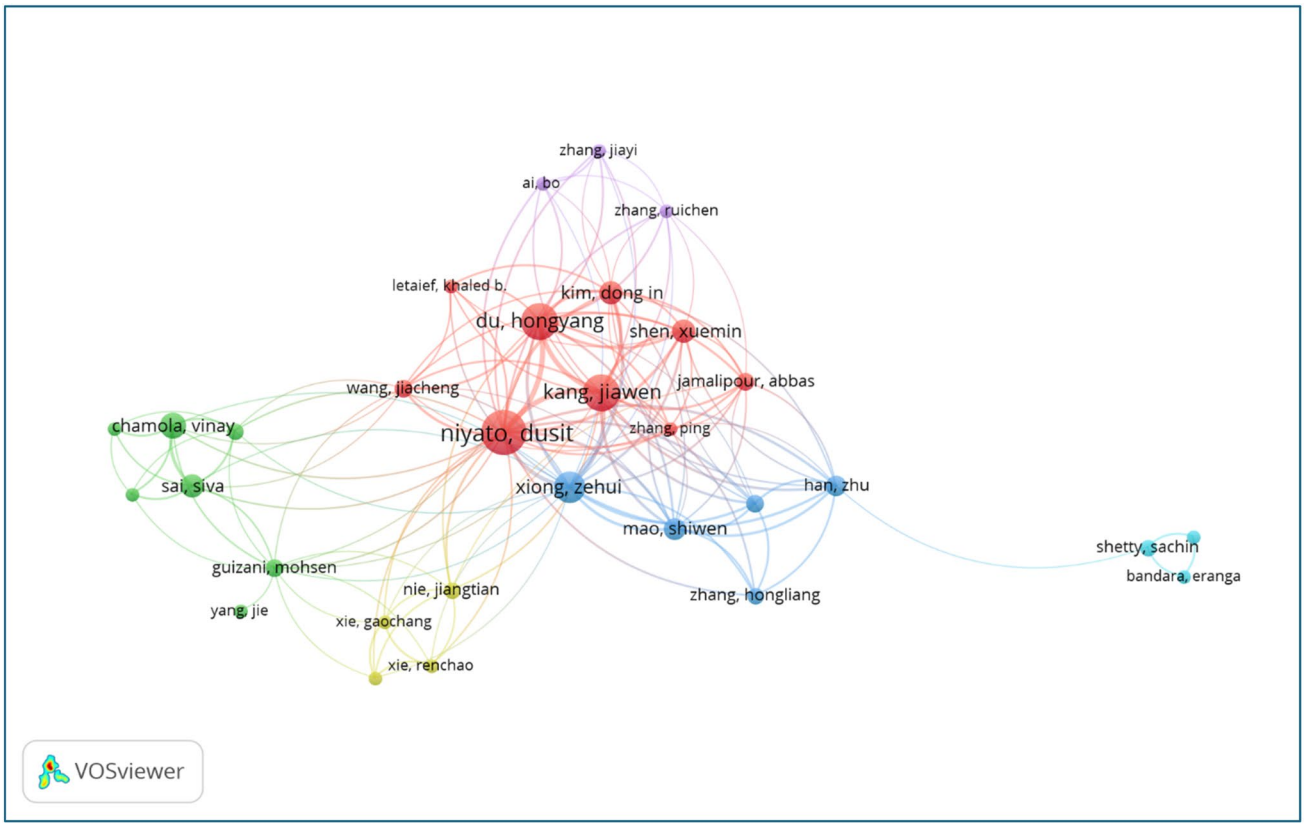


Fig. 9 Co-authorship network of authors in GenAI research

concentrated on themes; the red cluster denotes a core group of close-knit colleagues. The blue cluster notes writers in related fields of study. Smaller independent clusters with limited linkages arise from peripheral collaborations like Shetty and Bandara Eranga. The network emphasizes how working together will help to advance generative artificial intelligence research.

4.4 Countries contribution analysis

Ranked by the number of papers generated and citations obtained, Table 2 compares the top-performing nations' research on generative artificial intelligence, stressing their respective contributions and influence within the field.

With 775 documents, the United States leads under the "Based Upon Documents" criterion, much above other nations in research production. With 269 and 259 papers, respectively, India and China follow to highlight their increasing participation in generative artificial intelligence research. With 250 records, the United Kingdom ranks fourth; Australia comes in fifth with 134 documents. The "Based on Citations" standard verifies the research's influence and acknowledgment. With 4427 citations, the United States keeps its supremacy and supports its role as the most important contributor. Fascinatingly, although Hong Kong comes second with 1344 citations, its document count (84) is modest, suggesting a high

Table 2 Country-wise raking based on no. of documents produced and no. of citations obtained

Based upon documents				Based on citation			
Rank	Country	Documents	Citations	Rank	Country	Documents	Citations
1	United States	775	4427	1	United States	775	4427
2	India	269	460	2	Hong Kong	84	1344
3	China	259	842	3	Australia	134	1203
4	United Kingdom	250	1189	4	United Kingdom	250	1189
5	Australia	134	1203	5	Germany	132	880

average citation impact per publication. Australia, with 1203 citations, also has a great impact and ranks third. With 1189 citations, the United Kingdom ranks fourth with its document-based perspective. With 880 citations, Germany rounds out the top five regarding research influence. While underlining the great impact-to-production ratios of nations like Hong Kong and Australia, this dual-ranking approach emphasizes the supremacy of the United States in both research output and influence. The information shows the worldwide spread of generative artificial intelligence research and the diverse degrees of influence different countries apply.

With the United States as the biggest node, Fig. 10 shows a network visualization of national cooperation in generative artificial intelligence research. This suggests both its vital importance and great worldwide cooperation. India, China, Australia, Germany, and the United Kingdom are important partners. The network is split into red clusters for projects headed by India and the Middle East, green clusters linking European nations, including Germany and Spain, and blue clusters, including Australia and South Korea. Though they have smaller research output or limited cooperation, smaller nodes, including Nepal, Bangladesh, and the Philippines, are linked to bigger nodes. The thick network of links shows the rich degree of cooperation in generative artificial intelligence research, stressing the research ecosystem's multidisciplinary and worldwide connected character.

4.5 Keywords contribution analysis

The primary research topics are "GenAI" and "generative artificial intelligence," encompassing related concepts such as LLM, GPT, transformers, and computer vision, as depicted in Fig. 11. The network is segmented into color-coded clusters, emphasizing cybersecurity, artificial intelligence ethics, and accountability. The green cluster emphasizes technological innovations, whereas the orange and purple clusters investigate educational applications. The domain is profoundly interdisciplinary, featuring intersecting studies in technology, ethics, education, and practical applications. Emerging subjects encompass GPT-4, copilot, and data augmentation, whereas concepts such as policy, storytelling, and design thinking reflect wider societal and creative implications. The image accurately delineates the intellectual terrain of GenAI research, highlighting its varied yet interrelated issues. Figure 11 also indicates that GenAI, a notion characterized by

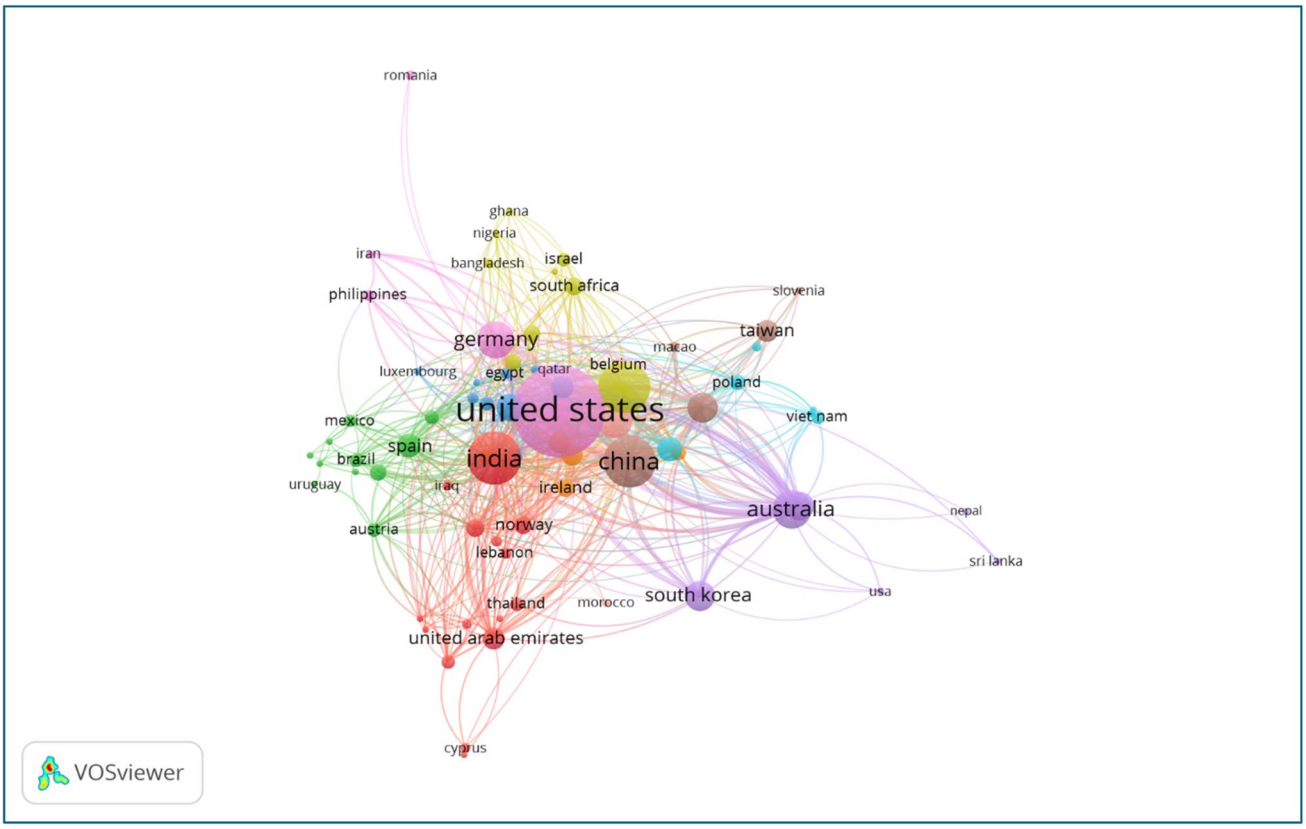


Fig. 10 Global research collaborations in GenAI: A VOSviewer network visualization highlighting the top contributing countries

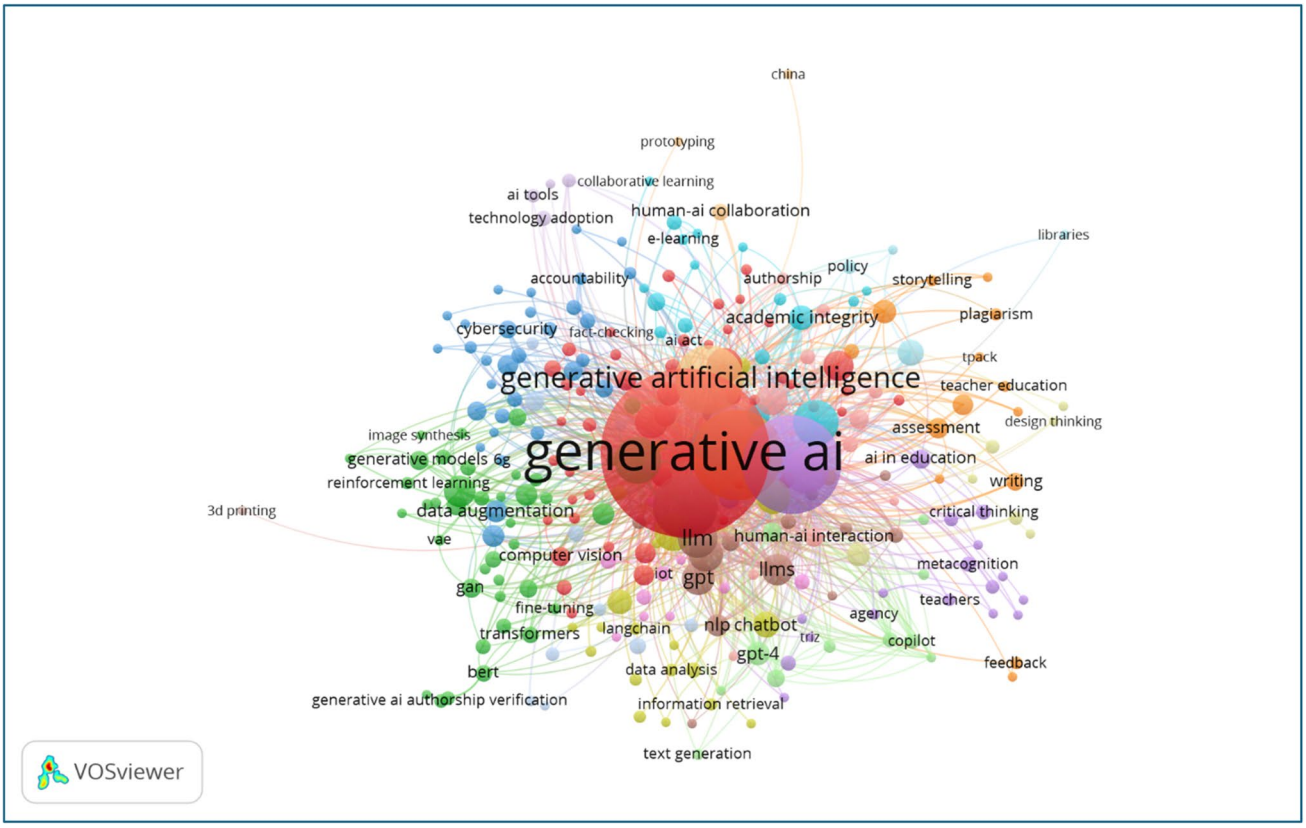


Fig. 11 Keywords network analysis

numerous occurrences and connections, is the primary focus in the field. This is a fundamental notion across several research disciplines, with ChatGPT positioned second, exhibiting 346 occurrences and a total link strength of 1010. Artificial Intelligence encompasses a wider scope, with 283 instances and a cumulative connection strength of 830. Large language models (LLMs) are pivotal subjects of interest, with machine learning as a fundamental methodology. Higher education and education are burgeoning fields where GenAI is utilized for pedagogical and learning contexts, tackling plagiarism and academic integrity. The ethical aspects of the field are being examined, focusing on responsibility, bias, and the responsible use of AI. The interrelated nature of research demonstrates considerable collaboration and interdisciplinary impact.

4.6 Topic modeling

The experiment highlighted keywords the authors and experts labeled, as represented in Table 3. These keywords serve as critical identifiers of the thematic focus and provide valuable insights into the core areas of discussion. Additionally, the authors curated a list of relevant articles outlining their contributions to the topic under study, ensuring a comprehensive understanding of the subject. Furthermore, in Table 4, the authors mapped the Sustainable Development Goals (SDGs) to the formulated clusters, establishing a clear connection between the identified clusters and their alignment with global sustainability objectives. This mapping provides a structured approach to analyzing the interplay between the research themes and the broader goals of sustainable development.

4.6.1 Topic I: Deep learning for image augmentation

The latest research studies in deep learning for image augmentation have been motivated by the necessity to improve model robustness and efficacy. Generative Adversarial Networks (GANs) and diffusion models are extensively employed to produce high-quality synthetic data, enhancing dataset variety and model generalization [75]. Adversarial training and attention mechanisms enhance accuracy by producing difficult instances and improving feature extraction. The

Table 3 Topic labels and their relevant contribution

Cluster no.	Keywords	Topic label	Relevant articles	Contribution percentage
1	image, dataset, method, network, performance, text, result, machine, accuracy, training, approach, propose, synthetic, adversarial, train, deep, high, feature, diffusion, augmentation	Deep Learning for Image Augmentation	[65]	98.86
2	author, risk, work, concern, article, copyright, legal, social, medium, content, ethical, issue, bias, public, make, impact, text, communication, large, question	Impact of Gen AI in Social Media Content creation and Ethical considerations	[66]	98.89
3	user, image, creative, text, creativity, content, interaction, prompt, create, author, art, visual, generation, video, creation, designer, process, owner, experience, game	Innovative Methods in Art and Content Generation	[67]	99.26
4	medical, health, patient, healthcare, clinical, llm, care, large, information, document, mental, result, improve, diagnostic, treatment, review, author, accuracy, report, knowledge	GenAI in Healthcare	[68]	84.12
5	ethical, explore, integration, application, educational, chapter, content, enhance, address, offer, include, provide, future, highlight, development, landscape, approach, framework, comprehensive, impact	Ethical Integration of Content Development in Educational Landscape	[69]	99.41
6	high, teacher, assessment, academic, finding, educational, impact, gai, teach, university, practice, educator, analysis, perceive, result, skill, teaching, literacy, learning, perception	Impact of GAI on Teaching and Learning Practices	[70]	99.79
7	business, industry, service, product, management, gai, innovation, customer, marketing, process, decision, quality, consumer, adoption, user, impact, organization, market, make, information	Impact of Gen AI on Business Innovation	[71]	80.79
8	development, process, application, software, analysis, review, engineering, knowledge, provide, future, practice, focus, large, llm, author, researcher, field, present, explore, literature	Future of LLM Applications in Research and Engineering	[72]	94.68
9	llm, prompt, large, code, engineering, task, text, process, result, provide, feedback, programming, user, software, generation, response, problem, approach, create, present	Gen AI in programming and software Development	[73]	99.43
10	network, gai, propose, security, ieee, communication, application, approach, attack, resource, service, cybersecurity, edge, enhance, user, threat, framework, digital, real, drive	Cybersecurity Threat Detection using Gen AI	[74]	98.9

Table 4 Mapping of topics to relevant SDGs

Topic number	Topic	Relevant SDGs	How It relates to SDGs
1	Deep Learning for Image Augmentation	SDG 3 (Good Health and Well-Being), SDG 15 (Life on Land)	It enhances medical imaging for better diagnostics (SDG 3) and supports environmental monitoring through satellite imagery (SDG 15) [90]
2	Impact of Gen AI in Social Media Content Creation and Ethical Considerations	SDG 16 (Peace, Justice, and Strong Institutions), SDG 10 (Reduced Inequalities)	Promotes transparency, combats misinformation (SDG 16), and democratizes content creation for marginalized communities (SDG 10) [91]
3	Innovative Methods in Art and Content Generation	SDG 4 (Quality Education), SDG 8 (Decent Work and Economic Growth)	Enhances educational content (SDG 4) and fosters economic growth in creative industries (SDG 8) [92]
4	GenAI in Healthcare	SDG 3 (Good Health and Well-Being), SDG 9 (Industry, Innovation, and Infrastructure)	Supports personalized medicine and resource optimization (SDG 3) and advances medical innovations (SDG 9) [81]
5	Ethical Integration of Content Development in Educational Landscape	SDG 4 (Quality Education), SDG 17 (Partnerships for the Goals)	Ensures access to ethical and personalized educational tools (SDG 4) through stakeholder collaborations (SDG 17) [93]
6	Impact of Gen AI on Teaching and Learning Practices	SDG 4 (Quality Education), SDG 10 (Reduced Inequalities)	Provides adaptive learning tools (SDG 4) and reduces educational disparities in underserved communities (SDG 10) [94]
7	Impact of Gen AI on Business Innovation	SDG 8 (Decent Work and Economic Growth), SDG 9 (Industry, Innovation, and Infrastructure)	Boosts entrepreneurship and operational efficiency (SDG 8) while fostering sustainable industrial innovation (SDG 9) [95]
8	Future of LLM Applications in Research and Engineering	SDG 9 (Industry, Innovation, and Infrastructure), SDG 17 (Partnerships for the Goals)	Accelerates innovation in engineering (SDG 9) and promotes collaborative research globally (SDG 17) [96]
9	Gen AI in Programming and Software Development	SDG 8 (Decent Work and Economic Growth), SDG 12 (Responsible Consumption and Production)	Improves productivity in software development (SDG 8) and optimizes resource usage for sustainable outcomes (SDG 12) [97]
19	Cybersecurity Threat Detection Using Gen AI	SDG 16 (Peace, Justice, and Strong Institutions), SDG 9 (Industry, Innovation, and Infrastructure)	Strengthens digital security and protects critical infrastructure (SDG 16, SDG 9) [98]

utilization of these technologies in sectors such as healthcare has experienced considerable expansion, particularly for training models with constrained medical datasets [76]. Multimodal augmentation, which integrates visuals with text or speech, is gaining prominence for activities like visual question answering. Future studies will likely enhance augmentation approaches for efficiency, mitigate biases in generated data, and refine small-scale dataset augmentation: cross-domain augmentation and the use of unsupervised learning methodologies present opportunities for more progress. Real-time augmentation methodologies could also transform domains such as autonomous driving and robotics [77]. Deep Learning for Image Augmentation significantly contributes to advancing many Sustainable Development Goals (SDGs) by improving the quality and applicability of visual data. SDG 3 (Good Health and Well-Being) enhances medical imaging modalities, including MRI, CT scans, and X-rays, facilitating precise diagnosis, early disease identification, and tailored therapy. SDG 15 (Life on Land) enhances environmental conservation by refining satellite imagery, facilitating improved monitoring of deforestation, wildlife habitats, and land-use alterations [78]. Furthermore, for SDG 9 (Industry, Innovation, and Infrastructure), augmented imaging enhances industrial applications, encompassing quality control in manufacturing, infrastructure surveillance, and autonomous systems.

4.6.2 Topic II: Impact of Gen AI on social media content creation and ethical considerations

Contemporary studies on the impact of GenAI (Gen AI) in social media content underline its ability to change communication, boost creativity, and create ethical conundrums [79]. Using AI-generated content in text, video, and image creation—which helps to enable fast content creation at scale—is a noteworthy achievement. However, when legal experts investigate how artificial intelligence affects intellectual property, ethical questions arise about copyright, authorship, and the proprietorship of AI-generated works. Studies have found prejudices in artificial intelligence models that can support stereotypes or produce negative content, fueling the desire for improved ethical guidelines to produce AI. Particularly when AI-generated content becomes increasingly indistinguishable from human-created content, growing concern emerges about the risks of misinformation, manipulation, and declining public trust. Future research will probably focus on improving techniques for spotting and reducing prejudices, guaranteeing openness and responsibility in artificial intelligence systems, and offering thorough legal frameworks to protect users and authors of material [80]. Moreover, examining the psychological and social consequences of AI-generated material on consumers and creating ethical guidelines and policies for artificial intelligence in media will be crucial. Future research should focus especially on addressing content screening and controlling artificial intelligence systems to prevent exploitation. GenAI can enhance SDG 16 (Peace, Justice, and Strong Institutions) and SDG 10 (Reduced Inequalities) by fostering transparency, accountability, and diverse representation in social media material. It can identify and counteract falsehoods, promoting a harmonious digital environment [81]. It also democratizes content creation, enabling disadvantaged persons to produce high-quality material without substantial resources. Nonetheless, ethical considerations such as data privacy, bias mitigation, and the prevention of AI-generated content misuse are essential for attaining these objectives sustainably and equitably. When applied judiciously, GenAI has the potential to revolutionize social media into a more inclusive and ethical environment.

4.6.3 Topic III: Innovative methods in art and content generation

Contemporary research on innovative art and content creation methodologies focuses on enhancing user engagement and creativity through artificial intelligence-driven tools [82]. By merely providing prompts, these tools, such as generative models, empower users to independently produce photographs, text, films, and other forms of content, thereby facilitating non-experts' involvement in artistic endeavors. Artificial intelligence facilitates the creation of original and tailored designs within the realm of visual and video content, thereby amalgamating the roles of artist and producer [82]. Particularly within gaming and virtual environments, this transformation fosters novel opportunities for collaboration between human designers and artificial intelligence, resulting in distinctive artistic expressions and interactive experiences. However, given that consumers generally possess minimal influence over the creative process of artificial intelligence, this situation raises significant concerns regarding authorship and proprietorship of AI-generated content. Particularly concerning the capacity of artificial intelligence to reproduce or replicate existing artworks, this raises significant inquiries regarding originality and copyright. Consequently, ethical considerations surrounding using artificial intelligence for content creation also emerge as a prominent issue. Future research will likely explore an increased variety of user interfaces, the integration of multimodal generative models, and strategies for safeguarding intellectual property within this evolving creative landscape [83].

Furthermore, forthcoming trends will be significantly influenced by an understanding of the psychological and emotive responses elicited by art generated through artificial intelligence and its impact on traditional art forms. The new art and content production techniques of GenAI support SDG 4 (Quality Education) and SDG 8 (Decent Work and Economic Growth). They improve education by developing interactive learning resources, promoting creativity, and automating monotonous work [84]. These technologies also create new economic prospects in media, advertising, and entertainment sectors, fostering sustainable development.

4.6.4 Topic IV: GenAI in healthcare

Research about generative artificial intelligence in the healthcare sector is progressively concentrating on enhancing diagnostic precision, personalizing treatment approaches, and optimizing the efficacy of patient care [85]. Artificial intelligence models, especially large language models (LLMs), are being employed to scrutinize extensive volumes of medical data, encompassing patient records, diagnostic reports, and clinical notes, to derive significant insights and facilitate informed decision-making. These models have demonstrated considerable potential in automating routine tasks, including medical documentation, literature reviews, and report generation, thereby substantially alleviating healthcare professionals' administrative burden. GenAI is increasingly employed to formulate personalized treatment plans by integrating various data sources, including patient histories, genomic information, and contemporary research findings, thereby augmenting the accuracy of medical interventions [86]. Mental health care has experienced significant advancements due to the implementation of artificial intelligence-driven tools for sentiment analysis, the early identification of disorders, and the provision of support via conversational agents. Nonetheless, ethical considerations about data privacy, model bias, and the transparency of artificial intelligence decision-making processes continue to pose considerable challenges. Future research is anticipated to concentrate on improving the interpretability of artificial intelligence models, safeguarding data security, and establishing frameworks for integrating AI tools into clinical practice, all while preserving human oversight.

Furthermore, an in-depth investigation into the application of generative artificial intelligence for real-time diagnostics and continuous patient monitoring is expected to propel advancements in preventive care and personalized medicine. GenAI in healthcare substantially advances SDG 3 (Good Health and Well-Being) and SDG 9 (Industry, Innovation, and Infrastructure) by improving diagnostic precision, facilitating medication discovery, and developing individualized treatment strategies [81]. It optimizes medical workflows and encourages innovation via AI-driven solutions for telemedicine, remote patient monitoring, and predictive analytics, thus establishing a robust healthcare infrastructure and enhancing overall health outcomes.

4.6.5 Topic V: Ethical integration of content development in educational landscape

Research concerning the ethical integration of content development within the educational sphere is progressively examining the challenges associated with incorporating technology, all while upholding principles of integrity and inclusivity. Artificial intelligence in content creation, particularly in generating educational materials, can enrich learning experiences by delivering personalized, diverse, and accessible content [87]. Ethical considerations must be meticulously addressed, such as biases inherent in artificial intelligence algorithms, content veracity, and data privacy safeguarding. Researchers are investigating frameworks that promote ethical content development, emphasizing inclusivity, equity, and harm mitigation. A holistic strategy for integrating AI-driven educational tools seeks to harmonize innovation with safeguarding students' rights and dignity [88]. The future of educational content development is expected to incorporate comprehensive ethical guidelines that will govern the creation and distribution of AI-generated materials, thereby ensuring equitable learning opportunities for all students. The influence of artificial intelligence on the teacher-student relationship, particularly concerning the function of human educators within a technology-driven context, constitutes a significant area of investigation. Future research will continue to underscore the significance of developing culturally sensitive, transparent content devoid of detrimental stereotypes, thus fostering ethical practices within the swiftly transforming educational environment. Incorporating ethical AI in education corresponds with SDG 4 and SDG 17 by fostering diversity, collaboration, and responsible innovation. GenAI produces customized, high-caliber instructional materials, guaranteeing accessibility for everyone [89]. Integrating ethical AI cultivates collaborations among educators, technology developers, governments, and organizations, tackling issues such as data privacy and equitable access and empowering learners worldwide.

4.6.6 Topic VI: impact of GAI on teaching and learning practices

GenAI (GAI) is changing teaching and learning methods and transforming educational environments, particularly in universities and academic environments. From automating administrative chores to providing individualized learning experiences, research shows how GAI is increasingly used to improve teaching strategies [99]. Learning is more customized and efficient when teachers use AI-driven tools to help with curriculum building, student assessments, and feedback, enabling real-time resources and adaptable learning paths; GAI also changes how students engage with knowledge, promoting critical thinking and literacy skills. Though some view AI as a useful tool to improve teaching and learning, others express worries about its effect on academic integrity, skill development, and the teacher-student relationship [100]. This changes the view of educators and students on the part of AI in education. Future studies will probably concentrate on delving deeper into these views and investigating how GAI may be successfully included in conventional teaching strategies without sacrificing the growth of vital abilities. Further research on how GAI might affect academic assessments, support lifelong learning, and correct educational access inequities is also much needed. Constant research will be essential to guarantee the ethical implementation of GAI as it develops and maximizes its possibilities in building a more fair and effective learning environment. GenAI significantly influences teaching and learning methods, closely aligning with SDG 4 (Quality Education) and SDG 10 (Reduced Inequalities) by transforming the delivery and accessibility of education. It can automate administrative duties, enabling educators to concentrate on instruction [94]. It diminishes educational inequalities by offering scalable solutions for marginalized populations, fostering inclusivity, and democratizing access to quality education. This methodology fosters continuous learning opportunities and mitigates disparities.

4.6.7 Topic VII: impact of Gen AI on business innovation

Generative artificial intelligence (Gen AI) is transforming various sectors by enabling organizations to streamline processes, enhance the quality of products and services, and make data-driven decisions. From marketing and customer service to product development and administration, generative artificial intelligence is progressively integrated into various corporate functions [101]. In marketing, artificial intelligence solutions empower organizations to create tailored customer experiences, optimize campaigns, and predict consumer behavior, significantly enhancing customer engagement and satisfaction. Similarly, GenAI facilitates the generation of novel concepts, expedites design iterations, and enhances product quality through rapid prototyping and testing, thereby augmenting the product innovation process [102]. AI-driven decision-making tools assist organizations in analyzing vast quantities of data, enabling swifter and more informed decision-making. Incorporating GenAI within organizations facilitates the automation of routine operations and provides previously inaccessible insights. This, in turn, contributes to the enhancement of operational efficiency, the reduction of costs, and the promotion of a competitive advantage. Businesses must, however, address challenges such as ethical considerations, data protection concerns, and the necessity for qualified personnel to manage AI systems effectively. Future research will likely examine the impact of artificial intelligence adoption on organizational culture, consumer relationships, and corporate models in the long term. The research will also focus on how companies can navigate the evolving regulatory landscape by utilizing GenAI to maintain competitiveness and foster continuous innovation across the market. GenAI (Gen AI) is revolutionizing company operations, enhancing economic development and creativity, and harmonizing with Sustainable Development Goals 8 and 9. It improves efficiency, productivity, and innovation by optimizing operations, minimizing costs, and developing high-value products and services. GenAI expedites technological progress, revolutionizing industries and infrastructure [103]. Its capacity to generate novel business models, augment cooperation, and elevate consumer experiences fortifies businesses' resilience and competitiveness. GenAI also creates employment in technology-driven industries, fostering equitable labor and economic prospects.

4.6.8 Topic VIII: future of LLM applications in research and engineering

The future of Large Language Models (LLMs) in research and engineering will significantly transform both disciplines by facilitating knowledge discovery, streamlining processes, and offering innovative approaches to addressing intricate challenges [104]. Large language models (LLMs) are employed to automate literature evaluations, analyze extensive datasets, and support researchers in synthesizing existing knowledge across various academic disciplines. This enables researchers to swiftly discern trends, identify voids, and extract insights, substantially expediting the research process. In engineering, large language models (LLMs) are significantly advancing software development by facilitating code generation, debugging processes, and the optimization of algorithms [105]. This enhancement in productivity concurrently

diminishes the likelihood of human error. As large language models (LLMs) continue to advance, their utilization in these domains will broaden to encompass more intricate functions, including aiding in designing complex systems, simulating engineering processes, and providing tailored recommendations for research trajectories. Future applications are expected to prioritize the enhancement of interpretability and explainability of outputs generated by large language models (LLMs) to foster trust and facilitate their integration into scientific methodologies. Furthermore, as the models expand in scale and functionality, there will be an increased focus on ethical considerations, including safeguarding data privacy, mitigating bias, and establishing frameworks to address the implications of artificial intelligence in research. Research endeavors will persist in investigating the integration of large language models (LLMs) into established procedures, with the ultimate objective of augmenting the efficiency and quality of research and engineering results [106]. Large Language Models (LLMs) possess the capacity to transform research and engineering through the automation of data processing, the expeditious testing of hypotheses, and the augmentation of cooperation. They can enhance design processes, refine simulations, and enable predictive maintenance, resulting in more intelligent and sustainable infrastructure [96]. Large Language Models (LLMs) promote international collaborations, enhance scientific progress and technological innovation, establish robust infrastructure and inclusive innovation ecosystems, and align with Sustainable Development Goal 9.

4.6.9 Topic IX: Gen AI in programming and software development

GenAI (Gen AI) is progressively altering programming and software development by improving productivity, automating repetitive tasks, and providing novel solutions to intricate issues. Large Language Models (LLMs) facilitate code snippet generation, automate debugging, and offer real-time feedback, thereby enhancing the efficiency of the development process [107]. These models allow developers to provide straightforward prompts or high-level task descriptions, resulting in AI-generated code that can be seamlessly integrated into software applications. LLMs contribute to code review by assessing code quality and efficiency and providing suggestions for enhancements. This enables software engineers to concentrate on the more creative and strategic dimensions of development. Gen AI's capacity to process extensive text-based information renders it useful for documentation creation, code explanation, and enhancing collaboration among team members [108]. Challenges persist, including quality and security assurance in AI-generated code, mitigating potential biases in training data, and managing AI response interpretability. Future research is expected to enhance AI models to manage more intricate programming tasks, augment their capacity to engage with various languages and frameworks, and ensure that developers can rely on and effectively implement AI within the software development lifecycle. GenAI (Gen AI) in programming and software development can enhance economic growth and promote sustainable practices per Sustainable Development Goals 8 and 12 [84]. It can enhance productivity, diminish coding duration, and decrease entry barriers, resulting in digital economies and technology-oriented employment. GenAI automates code generation, debugging, and optimization, fostering efficient and responsible software development. This diminishes resource utilization, hence endorsing concepts of responsible consumption and production. GenAI promotes sustainable software development, yielding business and environment advantages.

4.6.10 Topic X: cybersecurity threat detection using Gen AI

Innovative solutions for detecting and mitigating cyber threats in real-time are offered by GenAI (Gen AI), which greatly improves cybersecurity threat detection. Cyberattacks are becoming more complex, making it harder for traditional security systems to keep up. To tackle this, companies are turning to AI-driven solutions that can efficiently analyze massive volumes of network data [109]. Advanced frameworks for threat detection and response across diverse digital environments, ranging from enterprise networks to edge devices, are being proposed using Gen AI models [110]. These models can mimic potential attacks, spot trends in data flow, and find security holes in communication routes that could otherwise go unnoticed. Gen AI has several potential uses in cybersecurity. Still, one of the most important is the creation of adaptive systems that can learn and adapt over time to deal with new threats, improving response times and how resources are distributed. Artificial intelligence's (AI) capacity to interpret data in real-time also makes it better at warding off DDoS attacks, malware, and other illegal access [111]. Improving these AI models to detect threats better sooner and with more accuracy in increasingly complicated and dispersed network environments is likely to be the focus of future research. To ensure that AI models can work without endangering user privacy or the integrity of the network, there will also be a focus on developing strong systems that balance privacy and security. GenAI is essential for cybersecurity threat identification, improving digital infrastructure security, and advancing Sustainable Development

Goals 16 and 9. It identifies and alleviates dangers instantaneously, thwarting cyber-attacks, fraud, and data breaches [112]. The innovative nature of Gen AI enhances industrial security architecture, enabling firms to prosper securely. This cultivates trust, stability, and resilience, facilitating sustainable economic growth and technical progress. In summary, GenAI is crucial for upholding peace and justice in the digital realm.

5 Implications and limitations of the study

The implications and limitations of this study provide a comprehensive understanding of GenAI's transformative potential while acknowledging the constraints that may impact its broader adoption and application. By categorizing these aspects into theoretical implications, practical implications, and research limitations, the study offers valuable insights for researchers, industry professionals, and policymakers.

5.1 Theoretical implications

This study provides a methodical examination of developments in GenAI, offering significant insights for scholars and researchers. It contributes to the theoretical understanding of GenAI applications across healthcare, business, education, and cybersecurity sectors. The findings highlight how GenAI drives innovation, transforms conventional methods, and raises ethical considerations such as privacy, bias, and employment effects. Additionally, the study enriches the existing literature by providing a framework for future research, offering predictive insights into sector-specific trends, and emphasizing the need for responsible AI deployment.

5.2 Practical implications

The study offers industry professionals a structured approach to utilizing GenAI to enhance processes, deliver tailored solutions, and maintain a competitive edge. Policymakers can leverage the findings to develop regulations that balance innovation with ethical deployment, particularly in sensitive sectors such as healthcare and education. Organizations can utilize the study's insights to adopt GenAI securely and responsibly, addressing issues related to model transparency, data security, and workforce transformation. Key Practical Implications for Industry as follows:

- Healthcare organizations can adopt GenAI for automating clinical documentation, improving diagnostics, and mental health assessments.
- Educational institutions can deploy AI-driven learning tools to deliver personalized learning content while ensuring ethical safeguards.
- Business sectors can leverage GenAI for intelligent automation in customer service, marketing, and operations optimization.
- Cybersecurity firms can utilize GenAI models for real-time threat detection, anomaly identification, and system hardening.
- These implications provide actionable direction for aligning GenAI innovation with sector-specific needs.

5.3 Research limitations

Despite its contributions, the study has several limitations. It relies solely on publications from the Scopus database, which may exclude relevant insights from other sources. While effective, the study's focus on Latent Dirichlet Allocation (LDA) for topic modeling may overlook complex relationships within the data that alternative methods, such as Non-Negative Matrix Factorization (NMF) or neural topic models, might reveal. Furthermore, the temporal scope of the study may not fully capture the rapidly evolving landscape of GenAI, and sectoral biases may result in uneven coverage of applications across industries. Lastly, challenges related to the interpretability of AI models and incomplete data for 2025 may introduce uncertainties in forecasting future research trends.

6 Future research directions

Based on the findings and identified deficiencies in existing literature, numerous intriguing directions for future research arise. These guidelines seek to improve the comprehension, utilization, and ethical incorporation of GenAI across several sectors:

6.1 Expansion of data sources

Future research should include datasets from supplementary academic databases, including IEEE Xplore, Web of Science, Springer, and industry white papers. Coverage may be restricted, and bias may be introduced by exclusively utilizing Scopus. Emerging developments that have not yet been indexed in peer-reviewed sources can also be captured by including grey literature and preprints [113].

6.2 Adoption of advanced topic modeling techniques

Future research could investigate more complex techniques like Non-Negative Matrix Factorization (NMF), BERT or Neural Topic Models (NTM), even while Latent Dirichlet Allocation (LDA) provides scalability and interpretability. These methods might find more profound semantic structures and contextual links among research topics [114].

6.3 Sector-specific deep dives

It is imperative to conduct comprehensive research that concentrates on underrepresented disciplines, including governance, climate action, legal systems, and rural development. Nevertheless, the literature on the subject is still rather limited in its exploration of these topics, despite the potential for generative artificial intelligence to be immensely beneficial to these fields [115].

6.4 Ethical and regulatory framework development

Future studies should focus on the development and comparison of ethical frameworks, AI governance models, and policy guidelines. This includes examining legal implications, AI accountability, explainability, and the development of global standards for fair AI, particularly in education, media, and healthcare [116].

6.5 Societal impact assessment

Examining the psychological and social effects of AI-generated content is desperately important. Especially in global south environments, this entails investigating public opinion, trust in artificial intelligence systems, effect on employment, digital divide, and cultural sensitivity.

6.6 Real-time and multimodal systems

Future research ought to explore real-time and multimodal AI systems, including models that integrate text, image, and audio production. Robotics, virtual assistants, smart cities, and immersive education necessitate comprehensive understanding and assessment.

7 Conclusion

This research provides a thorough examination of current advancements in Generative Artificial Intelligence (AI), emphasizing its applications, upcoming research trends, and alignment with the United Nations Sustainable Development Goals (SDGs). The study employed Natural Language Processing (NLP) and Latent Dirichlet Allocation (LDA) to analyze 2162 research articles published from 2020 to 2025, identifying ten principal topic clusters across sectors

such as healthcare, education, business, and cybersecurity. The results indicate that GenAI possesses significant potential to automate processes, foster innovation, and provide tailored experiences across various disciplines. The report underscores significant issues, especially with ethical consequences like bias, data privacy, misinformation, and job displacement. This paper enhances the theoretical and practical comprehension of GenAI by providing a sector-specific analysis of its present effects, pinpointing research deficiencies, and suggesting systematic taxonomy and connection with the Sustainable Development Goals (SDGs). It emphasizes the significance of interdisciplinary collaboration, responsible AI implementation, and policy coherence to guarantee the fair and sustainable integration of GenAI technologies inside global systems.

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Declarations

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